

Tracking structural variants trajectories across climates with alignment-free allele-frequency estimation

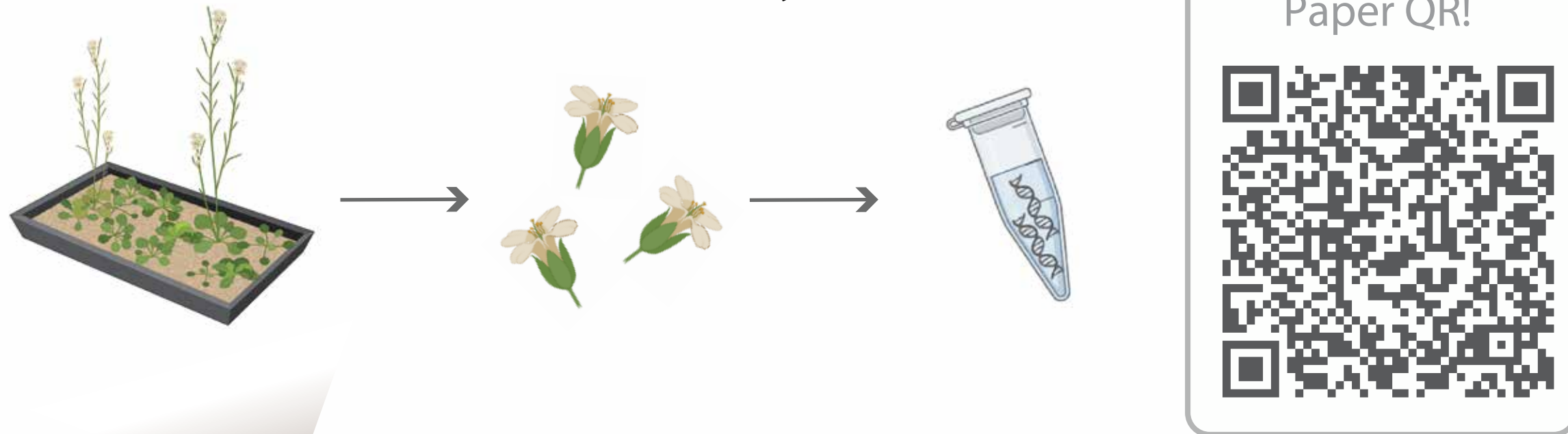
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The GrENE-net experiment

Synchronized outdoor evolution experiment with *A. thaliana* where an equal mixture of 231 accessions was planted in 12 replicates at 43 sites, and left to evolve under local climate for 3 years. Each year, surviving plants were sequenced as pools tracking allele-frequency change in real time.

2415 pools sampled and sequenced across the first 3 years



12 replicates

43 global locations running evolution experiments

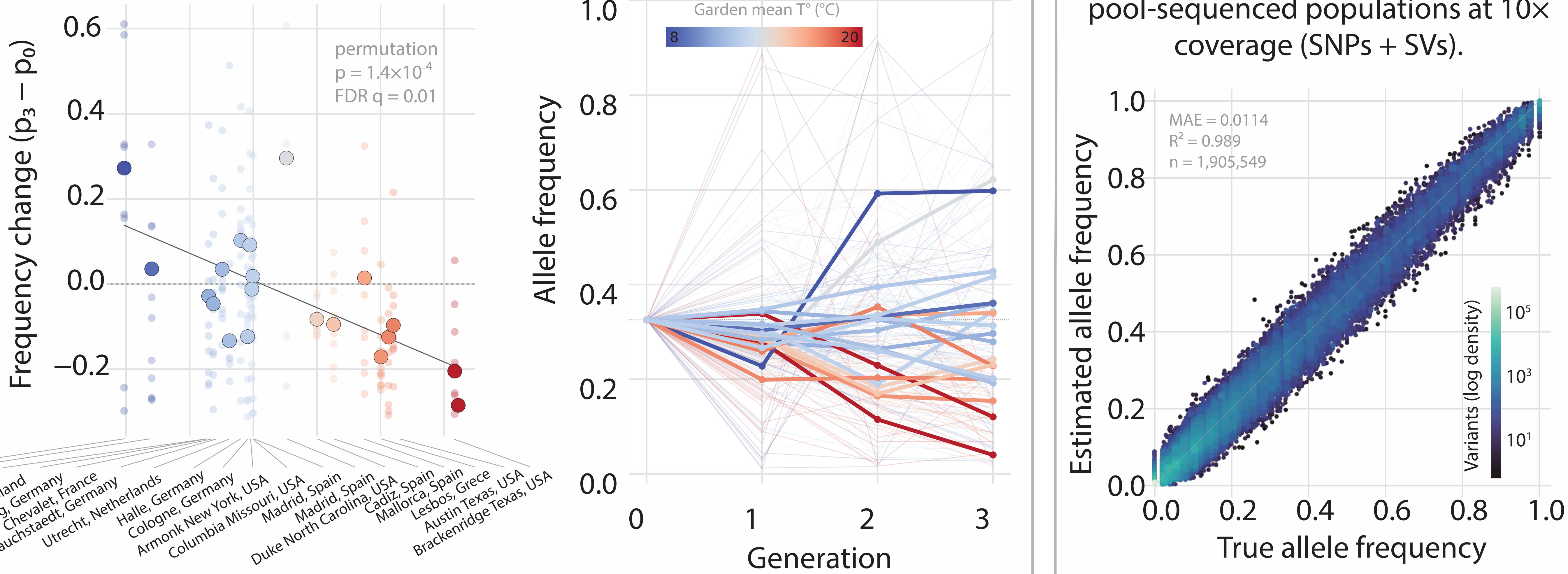
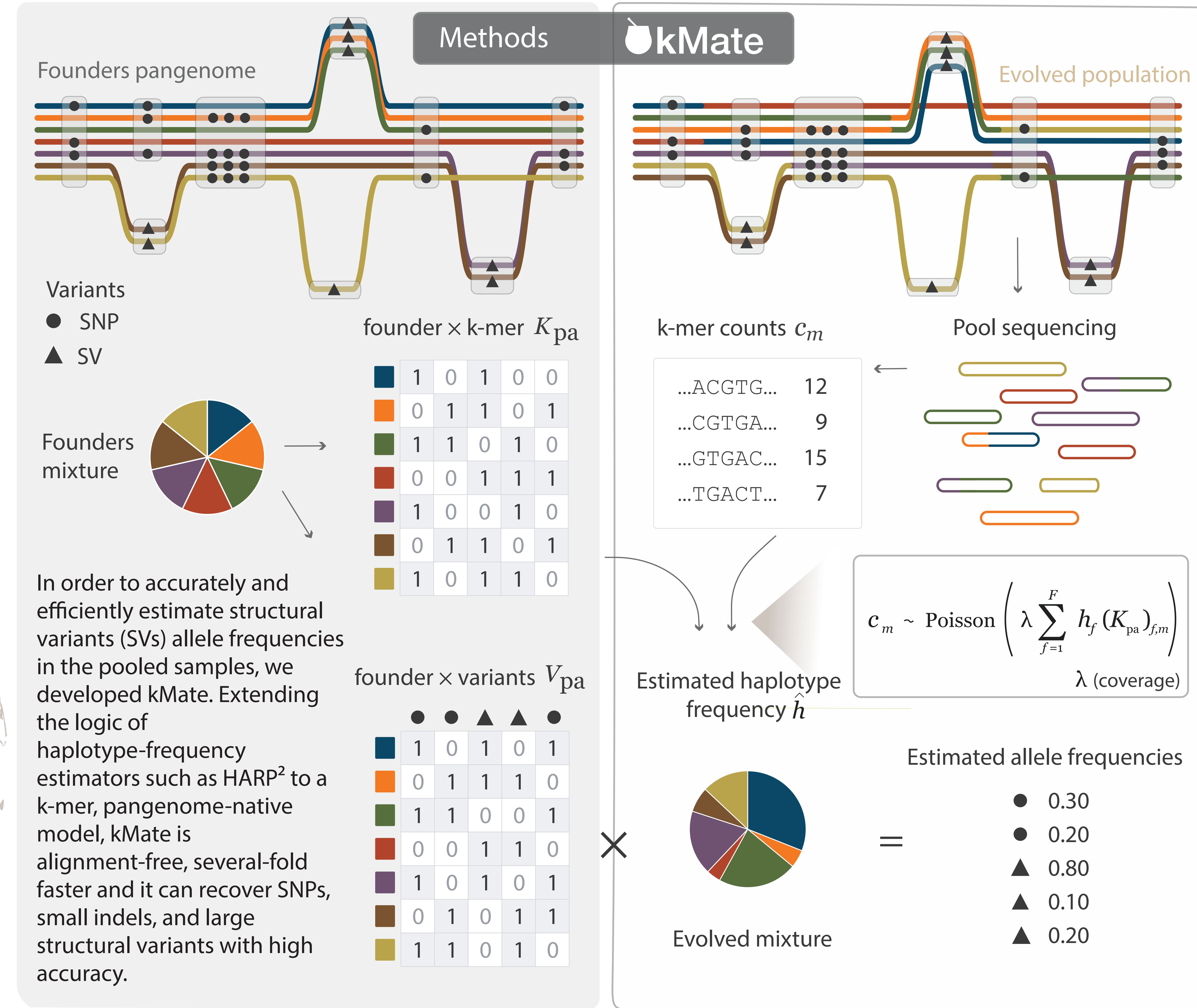
- Logistical fail
- No survival
- Survival 1 year
- Survival 2 year
- Survival 3 year
- Source accessions

Preliminary results

Once we obtained accurate SVs frequencies, we regressed per-garden SVs frequency change (Δp = gen 3 – founding) on gardens mean annual temperature.

Among significant peaks, we found a 181-bp insertion inside COR413-PM2 (COLD-REGULATED 413; AT3G50830) whose frequency rises in cold gardens and falls in warm ones.

Test kMate on your data!
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(1) Wu, Bellagio, Peng, Czech, Lin et al. Rapid adaptation and extinction in synchronized outdoor evolution experiments of Arabidopsis. Science 391 (2026). doi:10.1126/science.adz0777

(2) Kessner et al. Maximum likelihood estimation of frequencies of known haplotypes from pooled sequence data. Molecular Biology and Evolution 30, 1145–1158 (2013). doi:10.1093/molbev/mst016